When does an LLM autoresearch loop find real signal?

A controlled study on twelve biomedical prediction tasks

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Abstract

We applied LLM autoresearch loop [2] to 12 different biomedical prediction tasks from EEG, fMRI, microbiome, proteomic, and drug response data, and asked whether its improvements hold out on unseen data. Every final model was scored once on held-out subjects and, for three tasks, on external cohorts. We compared the loop with LLM proposals made without feedback with Optuna search [1], each allowed the same 20 development evaluations. Development scores improved in every run, but the held-out scores only improved in two thirds of them. On an average, the loop gain the AUROC of 0.05 over a standard baseline pipeline for each task more than the LLM without feedback, but this gain depended on a very small number of tasks. The loop clearly outperformed the Optuna on two tasks, the abnormal EEG and neonatal seizure detection, where its gains also replicated across independent runs. On the other tasks, part of the gain could be attributed to shortcuts rather than biological signal. When the site and demographic variables were provided, the loop built its models on them. In the TDBRAIN dataset, patients and controls could be separated using technical properties of the recording alone. Additionally, the loop's gain on psychiatric EEG did not transfer to external cohorts. These findings suggest that feedback is essential for an LLM to improve biomedical prediction pipelines, but the development scores alone overstate the loop can achieve. Gains from such loops should be confirmed on held-out, shortcut free and external data before they are interpreted as biological findings.

1 Introduction

2 Related work

3 Methods

Loop.

Tasks.

Shortcut-free protocol (v2).

Evaluation.

Same-budget baselines.

Statistics.

4 Results

4.1 Development gains are guaranteed; held-out gains are not

4.2 Feedback, not the LLM alone, produces the gains

4.3 Per-task and pooled statistics

4.4 Shortcuts and demographic confounds

4.5 In-distribution gains did not transfer to other cohorts

4.6 Variance, transduction and leakage

5 Discussion

When the loop helped.

When it did not.

Limitations.

6 Conclusion

Data and code availability

A Additional methods

Prompt and rules.

Shortcut-only models.

Age/sex confounds.

Transduction audit.

Leakage audit.

Pooled statistics.

B Disclosures

Decisions made after Eval results had been seen.

Unequal conditions.

Data-level choices.

C Dataset caveats

D Additional results

What the loop built.

E Loop trajectories

References

[1] Takuya Akiba, Shotaro Sano, Toshihiko Yanase, Takeru Ohta, and Masanori Koyama. Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pages 2623–2631, 2019.

[2] Andrej Karpathy. autoresearch. https://github.com/karpathy/autoresearch.